IEEE Advancing Technology for Humanity | IEEE Region 8 | IEEE Tunisia Section | IEEE Tunisia Section Student Activities Committee | IEEE Tunisia Section Technical Activities Committee | IEEE young professionals Tunisia Section | INSAT IEEE Student Branch

IEEE Entrepreneurship Tunisia Section | IEEE Women in Engineering WIE Tunisia Section WIE Affinity Group

TSYP 14

IEEE TUNISIAN STUDENT AND YOUNG PROFESSIONAL CONGRESS

IEEE TEMS Technology & Engineering Management Society | IEEE SSIT Society on Social Implications of Technology

Tunisia Section Joint Chapter

Women And Rural Agriculture Innovation Challenge

TSYP14 TECHNICAL CHALLENGE

SCOPE & TOPIC

This challenge invites engineering and entrepreneurship students to design technology-driven and business-driven solutions that strengthen the position of women in Tunisia's rural and peri-urban agricultural and artisanal economy.

The challenge focuses on:

- Technological automation to reduce manual labor and drudgery
- Digitalization & supply-chain structuring
- Occupational health & safety
- Direct market access for women-led ventures

Solutions should consider the real working conditions, safety and economic empowerment of women.

Challenge problem

Women agricultural workers, artisanal producers and micro-entrepreneurs play a major role in local production, including:

Olive oil harvesting • Essential oil distillation • Textile weaving • Agro-food processing

However, they face interconnected challenges:

- Low technological automation
- Fragmented and informal supply chains
- Occupational health risks
- Limited direct access to markets
- Reduced traceability and bargaining power
- Dependence on intermediaries

Addressing only one dimension has limited impact. Solutions should consider the interconnected nature of these challenges.

Challenge goal

Teams are expected to develop practical and deployable solutions that:

- Reduce Physical Drudgery
- Use technology to reduce manual effort and physical strain.
- Improve Occupational Safety
- Address health risks through protective, monitoring or ergonomic solutions.
- Digitalize Supply Chains
- Improve traceability and connections between producers, processors, distributors and buyers.
- Improve Market Access
- Enable women-led micro-enterprises to reach buyers directly through digital platforms, traceability or alternative distribution models.
- Promote Gender-Inclusive Engineering
- Design around the actual needs and working conditions of women.
- Build Deployable Solutions
- Develop prototypes and business concepts with realistic potential for deployment in rural Tunisian communities.

Thematic Tracks

TRACK 1: LOW TECHNOLOGICAL AUTOMATION

Develop technologies that reduce manual labor, physical effort and repetitive work.

TRACK 2: SUPPLY CHAIN FRAGMENTATION & LIMITED DIRECT MARKET ACCESS

Improve traceability, coordination, distribution and direct access to markets.

TRACK 3: OCCUPATIONAL HEALTH RISKS

Develop protective, monitoring, ergonomic or preventive solutions for safer working conditions.

Each team selects one primary track.

Challenge Instructions

WHO CAN PARTICIPATE?

- Teams of 3–4 members
- All student members must be IEEE WIE members
- All student members must be Entrepreneurship Community members
- At least 1 TEMS member
- At least 1 SSIT member

Phase 1: Preliminary Solution

Deadline: 05 November 2026

Teams must submit:

- 3-Minute Video: Show the progress of the challenge and present the technical solution.
  - Minimum editing preferred
  - Must remain anonymous
- Technical Report — Maximum 6 pages
  - Required structure:
    - i. Title
    - ii. Abstract — 150–250 words
    - iii. Keywords — 3–6 keywords
    - iv. Introduction
    - v. Related Work — if applicable
    - vi. Methodology / Materials and Methods
    - vii. Results and Discussion
    - viii. Conclusion and Future Work
    - ix. Acknowledgments — optional
    - x. References

Plagiarism checks will be applied, including internet sources and AI-generated content.

Phase 2: Solution Deployment

Deadline: 10 December 2026

Pre-selected teams must:

- Develop and deploy the technical solution prototype
- Submit a 4-minute presentation — PDF format
- Submit a 2-minute final demonstration video

Teams may also submit additional materials demonstrating the impact and creativity of their work, such as papers or posters.

Deliverables

PHASE 1

- 3-minute Pitching Video
- Technical Report max. 6 pages

PHASE 2

Finalists must provide:

- 4 minute Presentation PDF including proposed solution, technologies used, project architecture and Impact
- 2-minute Prototype Demo Video
- Deployed Technical Prototype
- GitHub Repository

Pitching

English | 4min Presentation | 2min Q&A

Scoring

TOTAL SCORE: 100 POINTS

FINAL SCORING SYSTEM — 100 POINTS

PHASE 1 — PRELIMINARY SOLUTION: 40 POINTS

- Problem Understanding & Social Relevance — 7 pts
- Innovation & Technical Solution — 10 pts
- Feasibility, Business Model & Impact Potential — 8 pts
- Technical Report — 10 pts
- Solution Pitch — 3-min Video — 5 pts

PHASE 2 — SOLUTION DEPLOYMENT: 60 POINTS

- Technical Solution & Demo — 20 pts
- Technologies & Engineering Quality — 10 pts
- Impact & Relevance — 10 pts
- Creativity, Consistency & Effectiveness — 10 pts
- Pitching, Presentation & Q&A — 10 pts

Important Dates

- Challenge Infosession: **16/09/2026**
- Phase 1 Submission Deadline: **05/11/2026**
- Final Submission Deadline: **10/12/2026**

Winners

3 WINNING TEAMS

Prizes: To be defined

Submission

Submit your project and show us what you've built.

Contact



Please contact us through the following email address wie@ieee.tn or tems@ieee.tn or entrepreneurship@ieee.tn for any inquires or clarifications.